

DEBUGGING AND OPTIMIZATION

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SIMATS Online Programming Lab
JAVA PROGRAMMING

NAGA TEJASWITHA
1923/1957RD

DEBUGGING AND OPTIMIZATION

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QUESTIONS

Q1
Debug Nested Exception Handling
10 marks

Q2
Debug Exception Propagation
10 marks

Q3
Debug Multiple Runtime Exceptions
10 marks

Q4
Debug Finally Execution
10 marks

Q5
Debug the NullPointerException
10 marks

5/5 answered

Q1 Debug Nested Exception Handling

10 marks

```
import java.util.Scanner;

public class Debug1 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = {20,40,60,80};

        int index = sc.nextInt();
        int divisor = sc.nextInt();

        try {
            int value = arr[index];
            try {
                int result = value/divisor;
                System.out.println(result);
            }
            catch (ArrayIndexOutOfBoundsException e) {
                System.out.println("Invalid Index");
            }
        }
        catch (ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}
```

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Q1 Debug Nested Exception Handling

10 marks

```
import java.util.Scanner;

public class Debug1 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = {20,40,60,80};

        int index = sc.nextInt();
        int divisor = sc.nextInt();

        try {
            int value = arr[index];
            try {
                int result = value/divisor;
                System.out.println(result);
            }
            catch (ArrayIndexOutOfBoundsException e) {
                System.out.println("Invalid Index");
            }
        }
        catch (ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}
```

Debugging Task
Identify why some exceptions are never caught.
Modify the exception handling so all runtime exceptions are handled correctly.

TEST CASE 1
Input 2 5
Expected Output: 12

TEST CASE 2
Input 5 2
Expected Output : Invalid Index

TEST CASE 3
Input 1 0
Expected Output : Division by Zero

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug1 {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int arr[] = {20, 40, 60, 80};
```

Test Input

2
10

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Q1 Debug Nested Exception Handling

10 marks

```
1 import java.util.Scanner;
2 public class Debug1 {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int arr[] = {20, 40, 60, 80};
6         int index = sc.nextInt();
7         int divisor = sc.nextInt();
8         try {
9             int value = arr[index];
10            try {
11                int result = value / divisor;
12                System.out.println(result);
13            }
14            catch (ArithmeticException e) {
15                System.out.println("Division by Zero");
16            }
17        }
18        catch (ArrayIndexOutOfBoundsException e) {
19            System.out.println("Invalid Index");
20        }
21    }
22 }
```

Test Input

2
10

Output

6

Test Cases

Test Case 1

Test Case 2

Test Case 3

10.00 / 10 marks

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QUESTIONS

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Debug Finally Execution
10 marks

Q5
Debug the NullPointerException
10 marks

5/5 answered

Q2 Debug Exception Propagation

10 marks

```
import java.util.Scanner;

class Calculator{

    static int divide(int a,int b){
        return a/b;
    }

    static void display(int x){
        System.out.println("Answer = "+x);
    }
}

public class Debug2{

    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt();
        int b=sc.nextInt();
        Calculator.display(Calculator.divide(a,b));
        System.out.println("Execution Completed");
    }
}
```

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```
System.out.println("Execution Completed");
}
}

Debugging Task
Prevent abnormal termination.
Ensure "Execution Completed" is always printed.
TEST CASE 1
INPUT: 20 5
OUTPUT: Answer = 4
Execution Completed
TEST CASE 2
INPUT: 50 0
OUTPUT: Cannot Divide by Zero
Execution Completed
TEST CASE 3
INPUT: 36 6
OUTPUT: Answer = 6
Execution Completed
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 class Calculator {
3     static int divide(int a, int b) {
4         return a / b;
5     }
6     static void display(int x) {
7         System.out.println("Answer = " + x);
8     }
9     public static void main(String[] args) {
10        Scanner sc = new Scanner(System.in);
11        int a = sc.nextInt();
12        int b = sc.nextInt();
13        try {
14            display(divide(a,b));
15        } catch (ArithmeticException e) {
16            System.out.println("Cannot Divide by Zero");
17        } finally {
18            System.out.println("Execution Completed");
19        }
20    }
21 }
```

Test Input

20 5

Output

Answer = 4
Execution Completed

Test Cases

Test Case 1

Test Case 2

Test Case 3

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Execution Completed

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 class Calculator {
3     static int divide(int a, int b) {
4         return a / b;
5     }
6     static void display(int x) {
7         System.out.println("Answer = " + x);
8     }
9     public static void main(String[] args) {
10        Scanner sc = new Scanner(System.in);
11        int a = sc.nextInt();
12        int b = sc.nextInt();
13        try {
14            display(divide(a,b));
15        } catch (ArithmeticException e) {
16            System.out.println("Cannot Divide by Zero");
17        } finally {
18            System.out.println("Execution Completed");
19        }
20    }
21 }
```

Test Input

20 5

Output

Answer = 4
Execution Completed

Test Cases

Test Case 1

Test Case 2

Test Case 3

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Debug Multiple Runtime Exceptions

10 marks

Q4

Debug Finally Execution

10 marks

Q5

Debug the NullPointerException

10 marks

5/5 answered

Q4 Debug Finally Execution

10 marks

```
public class Debug4 {
    static int calculate(int a,int b){

    try{
        return a/b;
    }
    finally{
        System.out.println("Resources Released");
    }
}

public static void main(String args[]){
    System.out.println(calculate(20,0));
}
}
```

Debugging Task

Prevent program termination.

Ensure the finally block executes while handling the exception correctly.

Test Case 1

INPUT: 20 5

OUTPUT: Resources Released

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5/5 answered

Prevent program termination.

Ensure the finally block executes while handling the exception correctly.

Test Case 1

INPUT: 20 5

OUTPUT: Resources Released

4

Test Case 2

20 0

Resources Released

Division by Zero

Test Case 3

81 9

Resources Released

9

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug4 {
3     static int calculate(int a, int b) {
4         try {
5             return a / b;
6         } finally {
7             System.out.println("Resources Released");
8         }
9     }
10    public static void main(String[] args) {
11        Scanner sc = new Scanner(System.in);
```

Test Input

20 5

Output

Resources Released
4

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★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug4 {
3     static int calculate(int a, int b) {
4         try {
5             return a / b;
6         } finally {
7             System.out.println("Resources Released");
8         }
9     }
10    public static void main(String[] args) {
11        Scanner sc = new Scanner(System.in);
12        int a = sc.nextInt();
13        int b = sc.nextInt();
14        try {
15            System.out.println(calculate(a, b));
16        } catch (ArithmeticException e) {
17            System.out.println("Division by Zero");
18        }
19    }
20 }
```

Test Input

20 5

Output

Resources Released
4

Test Cases

Test Case 1

Test Case 2

Test Case 3

10.00 / 10 marks

Submitted

